



Estimating group fixed effects in panel data with a binary dependent variable: How the LPM outperforms logistic regression in rare events data

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ABSTRACT

Estimating fixed effects models can be challenging with rare events data. Researchers often face difficult trade-offs when selecting between the Linear Probability Model (LPM), logistic regression with group intercepts and the conditional logit. In this paper, I survey these tradeoffs and argue that, in fact, the LPM with fixed effects produces more accurate estimates and predicted probabilities than maximum likelihood specifications when the dependent variable has less than 25 percent of ones. I use Monte Carlo simulations to show when the LPM with fixed effects should be preferred. I perform these simulations on common time-series cross-sectional (TSCS) data structures found in the literature as well as big data. This paper provides clarity around fixed effects models in TSCS data and a novel technique to identify which one to use as a function of the frequency of events in y .

1. Introduction

In this article I address common issues with fixed effects estimation in time-series cross-sectional (TSCS) data with a binary outcome variable. The three most common techniques used in political science to estimate fixed effects are the conditional logit (CL), the logit with dummies (LD), and the linear probability model (LPM) with fixed effects (LPMFE). Yet confusion remains regarding the benefits and drawbacks of these techniques as well as *when* to apply them. In this paper, I suggest that the decision on which method produces the most accurate estimates depends on the number of events in the dependent variable. I show that both maximum likelihood (ML) methods are best suited for dependent variables whose number of events is between 25 and 75 percent, provided the average number of observations per group is at least over 30. The LPMFE, on the other hand, performs much better with fewer than 25 percent of occurrences.¹ I also show that researchers can use the LD to produce predicted probabilities if certain conditions are met.

Whence the confusion with fixed effects? In TSCS data with a binary outcome variable, such models often require certain sacrifices that researchers are unwilling to make. With rare events data and fixed effects, a substantial portion of the sample may be lost in ML models since those groups without variation in the dependent variable are dropped. Coefficients are also notoriously inaccurate when large numbers of covariates are added to the model. The deficiencies of the LPM, on the other hand, are equally well-known: heteroskedasticity, predicted probabilities outside the 0–1 range, and the imposed linearity assumption. Because of the glaring deficiencies of both approaches with regards to fixed effects specifications, many researchers cite simplicity of interpretation as the reason for using

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¹ Or, trivially, more than 75 percent of ones. Henceforth I only make reference to rare events, but the logic applies to common events by trivial recoding.

the LPMFE or ‘best practices’ when using ML (Miller 2012, Harding and Stasavage, 2014).

While the reasoning is justified, I argue that this is the wrong conclusion. Fixed effects estimation can be difficult with a majority of available datasets, but the right steps can be taken to produce reliable results. In this paper, I seek to provide some clarity around existing fixed effects specifications and provide a new technique that helps researchers identify the fixed effects model that best suits their data according to the frequency of events in y . When observations per group are above 30, the ML and LPMFE models produce practically identical predicted probabilities *when the proportion of events in the sample is around 50 percent*. When the percentage of ones (or, trivially, zeros) is between 25 and 50, ML performs only slightly better than the LPMFE —primarily because nonlinear models are a bit more accurate at extreme values. Below 25 percent of events or non-events, however, the LPMFE produces predicted probabilities much closer to the observed probability for a majority of the distribution. Therefore, while the bias of ML with rare events is well known (King and Zeng 2001), bias occurs even when the number of events is relatively large (below 25 percent) when fixed effects are used. Indeed, the most important finding of this article is that, in cases of true rare events, i.e. when the number of ones is 1 percent or less, the LPMFE is the best method. Third, I show that logistic regression with dummies performs better than expected in big data analysis with a large number of both observations and groups. Despite this counterintuitive finding, researchers using big data should still use the LPM with fixed effects if the outcome variable is binary.

The structure of the paper is as follows. I first provide some theoretical reasons why using fixed effects is important and can help us identify false positives in our results. I follow this section with a discussion of the benefits and pitfalls of the three most common fixed effects models in the literature mentioned above —the LPMFE, the LD and the CL. I then use evidence from Monte Carlo experiments to establish that the LPM with fixed effects outperforms the LD and the CL in a majority of cases, especially in rare events data.

2. Fixed effects: theoretical background and notation

2.1. Why should we use fixed effects?

Comparisons across countries based on TSCS data, which is common in many social science fields, are particularly susceptible to omitted variable bias, as differences among countries are substantial and difficult to model. Fixed effects models provide a partial solution to this problem by invalidating time-invariant alternative explanations. Researchers who run the fixed effects equivalent of their model and find that their main coefficient of interest is *smaller* should be particularly alert to omitted variable bias (Allison 2009). Second, social science theories are usually based on interactions among key actors of different groups —governments, social classes and demographic groups, opposition parties, rebel groups, among others. Surprisingly, however, within-group variance is not often the focus of the models used in these fields of study even in cases where theories compare across many distinct groups, such as countries. Fixed effects models help us reach conclusions using within-group variance rather than simply exploiting variation between groups.

Questions remain about *how* and *when* to apply fixed effects. Should researchers use the CL, the LD, or the LPMFE? Theoretically speaking, since the dependent variable is limited, ML should be preferred over ordinary least squares. So one option is to add group dummy variables to a standard logistic model. However, this solution has two problems. First, if the number of group variables or intercepts increases with the sample size, ML’s asymptotic assumptions are violated and coefficients are biased. This is known as the ‘incidental parameters problem’ (Neyman and Scott 1948; see Section 3), which is particularly acute in dynamic models (Heckman 1981; Wooldridge 2002). Second, adding group dummies leads to systematic but small upward bias in the coefficients. This bias is the result of calculating many group intercepts — as is well known, logistic regression models are sensitive to the addition of many parameters (Beck 2018, 2020). The bias does not stem from the violation of any core assumptions and, in this sense, is less problematic.

A solution to both of these problems is to use the CL (Chamberlain 1980; see Heckman 1981). The CL produces a single coefficient that represents the log likelihood of observing y conditional on observing x for those groups that at least have one occurrence of x and variation in y . This makes the calculation more efficient and eliminates the incidental parameters problem. However, while the CL, at least in theory, might be the most appropriate method to perform fixed effects analysis in ML, it is especially complicated to use in practice. Predicted probabilities and confidence intervals often are difficult to compute and interpret since no fixed effects parameters are estimated. Predicted probabilities help us understand substantive significance, i.e. whether the effect of increases in economic equality on democratic transitions is politically meaningful (Achen 1982). The issues described above make this process difficult (see Wooldridge 2002). Recent work has focused on improving the performance of fixed effects logistic regression in rare events data, proposing innovative solutions such as Penalized Maximum Likelihood (Cook et al., 2018). Crisman-Cox (2020) shows that Correlated Random Effects models are preferred over the CL and the LD when the number of observations per group is small or events are rare.

The LPM with group intercepts is an attractive alternative to ML models (Beck 2018, 2020). While unconventional with limited dependent variables, the LPMFE presents two key advantages. One is that it allows groups with no variation in y to be included in the estimation, mitigating the issues with reduced sample size in the logistic model. Another benefit is ease of interpretation, both for the coefficients as well as the predictions. The main complication, however, is the linearity assumption, which can generate inaccurate probabilities at particular points of x even when the range of the predictions is accurate. Moreover, at extreme values close to 0 or 1, the LPMFE is likely to underperform. It may predict values over one or under zero or force linearity on a nonlinear relationship.

The argument in this article is that rare events exacerbate the shortcomings of logistic specifications with fixed effects, while linear probability models perform well despite data constraints. In a majority of datasets with a relatively large number of groups, rare events creates a separation between those groups with variation in y (*allV*) and those without (*noV*), as some have experienced the event in question and some have not. Oftentimes the difference between the number of groups with no variation in y and those with variation can be stark, which has deep theoretical and empirical implications for the estimation. In the next section, I explore these implications by comparing how each of the three models —Logit with dummies, CL and LPMFE— estimate their coefficients and calculate predicted

probabilities.

2.2. Three models to estimate fixed effects: the LPMFE, the conditional logit, and the Logit with group specific intercepts²

In logit specifications with fixed effects, groups without variation in the dependent variable do not enter the likelihood function. For instance, if a country never experienced a transition to democracy in the period under study, and thus the dependent variable is all zeroes, the coefficient for the group cannot be estimated due to lack of variation. Another way to see this is that the group dummy is perfectly correlated with the dependent variable, and the group is therefore dropped.³ The LPM specification with fixed effects does incorporate all groups into the estimation, as variation in y is not a requirement of OLS. The most interesting aspect of this specification is that β is a weighted average of two sets of groups, those countries with variation in the dependent variable (DV) and those which are all zeros (Beck 2018, 2020). For those groups where $y_{gi} = 0$, the estimate β is zero. For all others, it is the coefficient produced for all countries where there is variation in the DV. To see this process, let us take first the general equation for fixed effects and plug in a linear function such that

$$y_{gi} = x_{gi}\beta + \theta_g + \epsilon_{gi} \quad (1)$$

here, y_{gi} can be easily estimated using OLS and, in a context where the outcome variable is binary, the best estimate for $P(y_{gi} = 1)$ is $\hat{y}_{gi}$. We will use these predicted probabilities in the analysis to compare model performance. Note that the θ_g term represents the fixed effects parameters for the groups in the sample. These are known to induce bias in logistic regression coefficients under certain conditions, but linear probability model estimates are more robust to them (Beck 2018). The error term ϵ_{gi} satisfies the Gauss-Markov assumptions. Now, to see how OLS averages out the coefficients for groups with no variation (hereafter *all0*) in the DV and those with variation (*allV*), I show the equations for the OLS estimate for β_{all} and β_{allV} —for the entire sample and for only those that vary, respectively. I use the scalar case x , with all variables group mean centered, such that

$$\hat{\beta}_{all} = \frac{\sum (x_{allV,i})(y_{allV,i})}{\sum x_{all,i}^2} \quad (2)$$

$$\hat{\beta}_{allV} = \frac{\sum (x_{allV,i})(y_{allV,i})}{\sum x_{allV,i}^2} \quad (3)$$

the only difference in the two equations is that the denominator in (2) includes all values of x per group, while (3) includes only values for those groups in which there is variation in y . Given that the denominator includes values of the independent variable for groups *all0*, but the numerator does not, the coefficient β_{all} is a weighted average of the coefficient of both groups—it is as if the numerator had a second summation that returned zero. It is intuitive now that coefficients for OLS incorporate cases without variation in the DV and will be smaller than if those cases were dropped. The standard errors for the two equations are given by

$$SE_{\beta_{all}} = \sqrt{\frac{\hat{\sigma}_{all}^2}{\sum x_{all,i}^2}} ; \quad SE_{\beta_{allV}} = \sqrt{\frac{\hat{\sigma}_{allV}^2}{\sum x_{allV,i}^2}} \quad (4)$$

These equations show that, while the coefficients will be smaller because *all0* groups are taken into account, the standard errors will be smaller, as well. This is because the numerator in (4) left includes all observations, while the one in (4) right includes only those from the *allV* groups. Therefore, statistical significance will be affected to a lesser extent with the inclusion of *all0* groups. Substantively, however, the differences are large, as the predicted probabilities will certainly be smaller. This is, in my view, a strong suit of the LPMFE, as it does not statistically select on those countries that have variation in the DV. Some argue, however, that the LPMFE underestimates the coefficients by adding *all0* groups (Beck 2018, 2020). But I contend that this is akin to selecting on the dependent variable and the LPMFE allows us to avoid statistical selection bias. If China has not transitioned to democracy as inequality has increased because, let us imagine, repression has been particularly effective, why should we not include China in an analysis of the effects of repression on democratic transitions? This logic is even starker if we take the argument to its logical extreme: if only one country had transitioned to democracy in the period under study, we could not accept an argument that based its evidence on only one case. ‘Dropping’ cases, therefore, is logically difficult to justify, mostly when the ones that remain do so because our outcome of interest is observed.⁴ This is a more complex problem when events are rare and, therefore, the average is weighed down heavily by those groups without variation.

I will show how to best deal with that in the next section.

For logistic regression with group dummies, the coefficients are given by,

² This discussion is based fundamentally on Angrist and Pischke (2008), Beck (2018, 2020) and Hosmer et al. (2013).

³ Some statistical software packages, such as Stata, automatically drop the unit effect estimates while some others, such as glm in R, do not. However, the unit effect estimates tend to negative infinity and, therefore, they are not retained in the log-likelihood.

⁴ This is, indeed, a philosophical difference that does not affect the results or the application of the results of this paper. It is relevant, however, because in rare events *many* groups are usually dropped for lack of variation in the DV.

$$P(y_{gi} = 1) = \frac{1}{1 + e^{-(x_{gi}\beta + \theta_g)}}, \quad (5)$$

where the probability is estimated per group first, and therefore it cannot be computed for those groups that do not vary in y . The average coefficient for all groups will therefore include only *allV* groups. In rare events data, this estimation process complicates the reliability of the results, as a majority of groups drops from the sample, leading to overestimated coefficients, small samples and, consequently, larger standard errors.

The last method is another ML technique, the conditional logit, which uses the true conditional likelihood of observing an outcome given a set of parameters provided at least one event is observed (Chamberlain 1980). Rather than providing a group specific intercept for all groups in the regression, the conditional logit assumes the fixed effect to be zero and computes the conditional probability of observing an event. No intercept is therefore provided in the model. Let g denote groups, T_g the total number of observations for each group g , and y_{gt} is the value of the dependent variable, which takes on values of 0 and 1. Then,

$$\sum_{t=1}^{T_g} Y_{gt} = k_{1g}, \quad (6)$$

where k_{1g} is the number of observed events in group g . What the conditional logit then calculates is the probability of y_g conditional on $\sum_{t=1}^{T_g} Y_{gt} = k_{1g}$ such that

$$P\left(Y_g \mid \sum_{t=1}^{T_g} Y_{gt} = k_{1g}\right) = \frac{\exp(\sum_{t=1}^{T_g} Y_{gt} \text{it} \beta)}{\sum_{d_g \in S_g} \exp(\sum_{t=1}^{T_g} d_{it} \text{it} \beta)}, \quad (7)$$

where d_{gi} is 0 or 1 with $\sum_{t=1}^{T_g} d_{it} = k_{1i}$, and the term S_g represents all possible combinations of ones and zeros per group (Hosmer et al. 2013, eq. (7).4). Without going into more estimation details, what can be seen from this equation is that the conditional logit does not estimate a constant or different group specific intercepts, making it more robust to incidental parameters such as country dummies. In doing so, however, it also complicates the estimation of predicted probabilities, upon which political scientists rely for interpreting the substantive significance of statistical results.

Why, then, can we not just use the LD? The answer is that most applied researchers, in fact, are able to use the LD because the asymptotics of a majority of models are not in the group. Others who work with individual-level survey data, however, may be violating the incidental parameters problem.

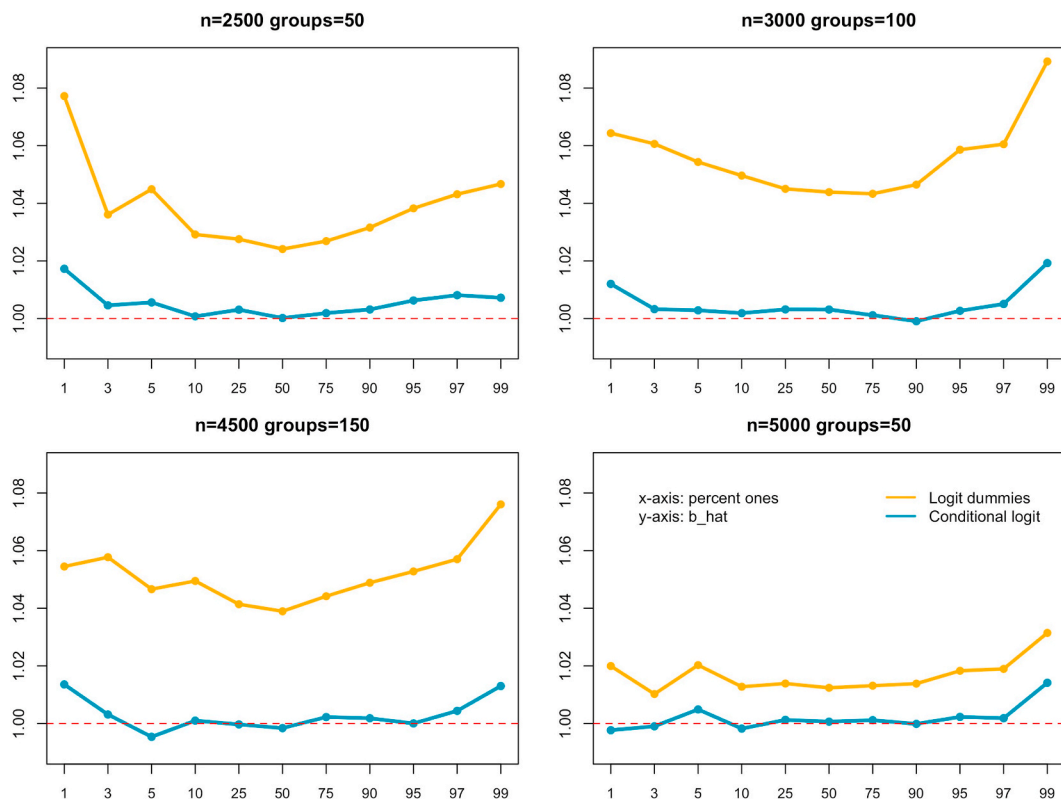


Fig. 1. Conditional logit coefficients versus logit with dummies.

3. The incidental parameters problem and the conditional logit

The incidental parameters problem, identified by [Neyman and Scott \(1948\)](#), is a central issue to ML estimation with fixed effects: when the number of parameters to be added grows with the number of observations, the estimates can no longer converge to the population parameter with shrinking standard errors as the sample size increases. Therefore, the asymptotic assumption at the core of maximum likelihood can no longer be maintained. [Andersen \(1973\)](#) and [Chamberlain \(1980\)](#) showed that using a conditional ML estimator solved the issue by removing the fixed effects parameters from the estimation.

The key question to ask, then, is whether the types of data that this paper relates to are susceptible to the incidental parameters problem. That is, whether the number of observations grows with the number of parameters as we approach infinity. The answer is no. Most datasets have their asymptotics at the group level, and the most common groups are countries, districts or regions in comparative politics or congresses, states and districts in American politics. Such groups in TSCS data cannot grow too large to generate a problem, and it is rather more likely that the estimation is ‘fixed’ in both time and group. The complication emerges with individual-level data, such as survey data, and we estimate a model using fixed effects at the *individual* level. If the asymptotics are in the individual, then we do have an incidental parameters problem and are forced into conditional estimation in ML.

Therefore, if the asymptotics are at the group level, and the group is fixed, researchers can use the LD for estimation, and one can make a strong case that there is no incidental parameters problem. However, special attention needs to be paid to the fact that the logit with dummies produces estimates that are slightly overstated. This bias stems from the fact that the logit with dummies actually estimates all the fixed effects in the model, that is, a coefficient is produced for each of the ‘incidental’ group parameters. Since, as is well known, ML is susceptible to large numbers of independent variables, adding a large number of groups introduces bias, but this bias is generally small and in a clear direction, as compared to the conditional logit.

[Fig. 1](#) shows the bias of the logit with dummies and the accuracy of the conditional logit. It draws from evidence obtained through four of the Monte Carlo simulations performed for the analysis section of this paper. The title of each plot shows the dataset structure used in each simulation—sample size and the number of groups in each dataset. Eleven simulations are performed by dataset, each one increasing in the proportion of events observed in y in the data. They range from rare events (1 percent) to highly common events (99 percent), and constitute the x -axis. In each of these simulations, I obtained average coefficients for the LD and the CL (blue and gold lines). 1 is the value of the β_1 coefficient in the true model in all the simulations. We can see how even with a large number of observations per group (100; bottom right), the logit with dummies produces consistently biased estimates around 1.02 when the true value of the parameter is 1. When the data is rare events (the points furthest to the left and right of the lines in each plot) the coefficients become more biased, as expected. The CL, on the other hand, produces unbiased coefficients throughout, except when the data is true rare events, even though the bias is always smaller than in the LD. The plot with 3000 observations and 100 groups (top right) shows that the tendency is exacerbated when the number of observations per group decreases to around 30. Here the logit with dummies produces biased coefficients around the 1.05 mark when events are not rare, and between 1.06 and 1.08 when they are. The conditional logit, on the other hand, is much more reliable, with the exception of true rare events, where coefficients tend to also be biased upward.

The problem, however, with the conditional logit lies in calculating predicted probabilities. To do so, one must *assume* that the fixed effect is zero. As will be shown later, doing this can lead to unrealistic probabilities in certain applications. Also, since it does not estimate a baseline, the model’s prediction when x is at its lowest point is often overstated in rare events. Thus, even if conditional logit coefficients are better estimates, they are a hindrance for substantive significance (see [Achen 1982](#)). I contend that the LPMFE, when events in y are infrequent, solves many of these issues and is the appropriate tool for applied researchers.

4. The problems with fixed effects regression using ML and the LPMFE

Multiple issues arise when we estimate fixed effects with ML. First is the loss of data that originates in ML estimation for those groups that have no variation in the DV. As shown in Section 2, no ML estimate can be produced for those groups with no variation in the DV. These groups are then dropped from the final sample. Conversely, the LPMFE produces 0 for such groups and the observations are kept. In rare events, the loss of data in ML can represent a problem, as a large portion of the data is often lost. Relatively large samples with over 30 observations per group can be turned into small samples of under one thousand observations, where we know ML estimates are biased.

A related issue, in the author’s view, is the implicit statistical selection bias in choosing only those cases that have variation in the dependent variable. If, let us say, a country with very high levels of inequality *never* transitioned to democracy, it would be removed from the estimation in ML models. Yet the substantive importance of this one country cannot be questioned. I refer to this process as ‘statistical’ selection bias. Knowingly eliminating groups that *are at risk of experiencing an event* but which did not actually experience it in the data is conceptually problematic. As [King and Zeng \(2001\)](#) show, for extreme cases of rare events, one may justify this statistical selection on the DV on certain solid theoretical grounds. But, I argue, doing so simply for the reason that a group did not experience an event is weak. Again, this is not an issue with the LPMFE, which constructs an average for those countries with no variation in the DV, yielding zero, and a separate estimate for those that do vary.

Moreover, the accuracy of the coefficients decreases in ML when a large number of covariates is added to the model. Even with relatively large ratios of observations per group (which usually need to be at least above 30), ML coefficients become increasingly biased as we add covariates to the model ([Beck 2018, 2020](#)). Another issue, which we saw in Section 3, is the difficulty in calculating predicted probabilities and, more importantly, marginal effects with Chamberlain’s conditional logit. Assessing substantive significance can be complex for researchers that do not possess ideal datasets—very large samples with a lot of observations per group and

strong variation in the DV.

The LPMFE is not without its own issues. First, as it fits a linear model on a probability space, it may produce nonsensical predictions over 1 and under 0. Second, it explicitly violates the heteroskedasticity assumption, since errors are by definition not randomly distributed across observations. Third, it issues predictions linearly, which means that predicted probabilities at different points may be understated or overstated if the relationship is indeed nonlinear –and we may never know. There are some interesting solutions to get around these problems: we can simply assume that nonsensical probabilities are simply close to 0 or 1, and we can use a Huber-Whit correction for heteroskedasticity (Huber 1967; White 1980). The linearity assumption is certainly the hardest to ‘correct’, but I suggest that we can confidently apply the LPMFE and its linearity assumption *as a function of the occurrence of events in the DV*. If events are rare or highly common, or if the number of ones is around 50 percent, the LPM will map on linearly to parts of the logistic CDF and produce reliable probabilities. I now proceed to the analysis section, where this argument should become clearer.⁵ Still, in a field where omitted variable bias is a salient problem, fixed effects models provide a better alternative to model overspecification and the inevitability of type-I errors. The second objection is not completely true. Rather, fixed effects exploits *within* variation, and I argue that this is often a better test of our theories, which usually revolve around rational choice explanations of domestic phenomena. Cross-national differences are exploited, but most theories use a logic that applies to changes within countries, not differences between them.

5. Analysis

I use Monte Carlo simulations to compare the performance of the LPMFE and the LD in calculating predicted probabilities. I do not use the CL here, for two reasons. First, because it does not estimate the fixed effect or a constant term and its predicted probabilities are generally inaccurate, a problem which is exacerbated with rare events data. Therefore, it is difficult to interpret the substantive significance of our results using the CL as our main model, but it serves as a useful robustness check given its accurate coefficients. Second, because in substantive comparative research the incidental parameters problem is not usually an issue, the LD can be used without incurring in this violation – even though, as I showed earlier, the coefficients will be increasingly biased as events in y become less frequent. I will show that, provided a large number of observations per group (around 100), the ML and the LPMFE provide similar results when the number of observed events in the data (the proportion of ones) is between 25 and 75 percent. Below or above this threshold, the LPMFE approximates the observed probability with increasing accuracy. At rare events (1% of ones or less) or highly frequent events (99% of ones or more), the argument will be stronger, that is, that the LPMFE should be the model of choice among the alternatives presented here, as it yields the most consistently accurate predictions.⁶ Lastly, while the performance of the LD improves with big data, which is counterintuitive, the LPMFE is still preferred with big data when events are rare.

5.1. Logit versus the LPMFE: Monte Carlo results

In this section, I introduce evidence from Monte Carlo simulations to ascertain the accuracy of the LPMFE and logit in calculating predicted probabilities using group fixed effects. The simulations test these two modeling techniques, LD and LPMFE, along three dimensions: (1) the distribution of the outcome variable, i.e. how frequent observed events are, (2) the number of groups or incidental parameters added to the model, and (3) the number of total observations and observations per group.

5.2. Simulation design

Variables Y^* and Y are generated from a true model

$$Y^* = 1.5 + 1 * X_i + \beta g * G_i + \mu_i,$$

$$Y = 1 \text{ if } Y^* > q_j$$

$$Y = 0 \text{ if } Y^* < q_j,$$

where X_i is a randomly generated variable with mean 0 and a standard deviation of 1. I include a common intercept of 1.5 and fixed effects terms represented by G accompanied by their own set of β coefficients, which are also fixed.

Within each group⁷ Errors μ_i are distributed logistically in the simulations. q_j is a vector of j quartiles of Y^* , each of which produces exactly the same percentage of ones in the data as the percentile in separate simulations. The vector used establishes breakpoints at six

⁵ Two other ‘philosophical’ objections to fixed effects need to be addressed. First is the claim that fixed effects is an overly blunt solution to control for unobservable heterogeneity. The second is that it is not sufficiently grounded in theory. There is some truth to the first objection, in that the fixed effects model removes all time-invariant heterogeneity and thus may sharply decrease variance.

⁶ Note that other approaches exist to deal with within- and between-group heterogeneity in TSCS data, such as random effects models, correlated random effects specifications, penalized logistic regression, etc. I selected the models in this paper due to their wide use in many fields within the social sciences, including economics, political science and sociology and to the fact that they often require the researcher to make a direct choice among them.

⁷ I draw a set of group β from one single draw of a uniform distribution. To ensure the robustness of the simulation set-up, I have replicated the simulation set-up using a de-meaned x_1 and omitting the group intercepts in the true model.

different percentiles: the 1st, 3rd, 5th, 10th, 25th, and 50th. This creates exactly one, three, five, ten, twenty-five and fifty percent of events in each dataset, respectively. For the big data tests, I run the analysis only at true rare events (the 1st percentile, with only 1 percent of observed events in the data). Thus, a total of six sets of simulations are run for each dataset structure presented below in Table 1 – except for datasets with 100,000 observations, with one set of simulations at rare events. Each of these produces results at different levels of event occurrence. The dataset structures chosen in this paper are intended to (1) approximate those usually found in comparative politics research and (2) reflect what would be ideal datasets in terms of total observations and observations per group. I report all the dataset structures used in the simulations in Table 1. They vary between 1050 and 100,000 total observations and between 20 and 4000 groups, yielding balanced panels with varying numbers of observations per group. From a brief survey of the literature, I consider that most time-series cross-section datasets in comparative politics vary between around 1000 and 7500 observations. Some datasets may be larger than 7500 observations, but these are rare. Similarly, datasets below 1000 observations exist, but logistic models are known to be biased below this threshold –and this bias will only increase if fixed effects are used. I also include two datasets, both of which have 100,000 observations, which are intended to produce results for researchers who use big data with binary response models and who may be uncertain about using fixed effects specifications. One of these datasets has 1000 groups and 100 observations per group, while the other has 4000 groups and just 25 observations per group. Again, in the case of the LD, the results hold only when the asymptotics are in N , not in the group. Again, this is not an issue with the LPMFE.

For each of these panels, the first simulation performs the estimation with rare events data (1 percent of ones) and the last one with full variation in y (50 percent of ones). Thus, a total of 110 simulations are run –11 simulations for each of the 10 ‘common’ dataset structures and one each for the big data structures. The one where the median is used provides the maximum amount of variation in the dependent variable and, therefore, is the one where the logit model should most closely approximate the observed probability. Comparing the coefficients and the predicted probabilities from these simulations sets the basis for the contrast between the LPMFE and the logit models in rare events data versus more common types of events.⁸

Each simulation of 1000 iterations produces, first, coefficients for the LD, CL and LPMFE.⁹ With the LD and LPMFE estimates, I then calculate predicted probabilities using the observed value approach (Hanmer and Kalkan 2013) and store them for each simulation. Since applied researchers rely on predicted probabilities to assess substantive significance, these predicted probabilities will be the focus of the comparison –rather than the coefficients themselves. I then plot these predictions for the LD and the LPMFE at different levels of event occurrence in the DV. These comparisons will be the central source of evidence in this paper. To gauge which model provides a better estimate in each simulation, I calculate the observed probability of y at different levels of x .¹⁰

6. Results from Monte Carlo experiments

In Figs. 2, 4 and 5, I report a representative set of results from the simulations described above. These results are from simulations using three dataset structures: 2000 observations and 20 groups, 3000 observations and 100 groups, and 2000 observations and 100 groups. These are common dataset structures in comparative politics and cover three different important scenarios. The first (2000/20) has 100 observations per group and adds a smaller amount of incidental parameters, 20. The second data structure has a bigger overall sample of 3000 observations and a borderline number of observations per group of 30. The last one (2000/100) yields 20 observations per group, and any average group size below 30 is known to introduce bias in ML coefficients (Beck 2018). Therefore, the first model should have the least amount of bias in the estimation, since there are 100 observations per group.

Each of the graphs shows the results for the LD and LPMFE models as well as the observed probability of y at each level of x . The x -axes in all graphs represent values for an x variable generated randomly with mean 0 and a standard deviation of 1. The y -axes are predicted probabilities. The blue line represents LPMFE predictions and the red line, LD probabilities. The black line plots the observed probability of each event occurring at different levels of x . It is calculated as the average value of y at different points of x in the data. The observed probability is not intended as a ‘true’ probability from which we can infer an exact measure of bias, but as a reference point to that any model’s predictions should approximate. The line is not produced by a specific model and thus serves as an impartial arbiter for visual comparisons of the predictions.¹¹

LPM vs. LD with 100 observations per group (2000 N ; 20 G).

Fig. 2 reports the results for a dataset with 100 observations per group. The subplots (a) through (f) display the predicted

⁸ Note that the simulations in this article only include one linear term and do not deal with other model specifications such as quadratic terms, interactions, splines, etc. Theoretically, however, one would expect the LPMFE to perform better with these specifications as well, as they all require adding more parameters to the model, which introduces bias in the logistic models. More empirical work is required on this front to ascertain if these expectations bear out in practice.

⁹ The CL coefficients were already shown in Section 3.

¹⁰ In the Appendix, I provide the following robustness checks by modifying the data generating process in two important ways. First, I change the true value of the parameter of interest in the simulations. In the main results section of the paper, I use a true value of 1. For robustness, I use true values of 0.5 and 2 using two dataset structures, one known to be biased (2000 observations and 100 groups) and one unbiased (2000 observations and 20 groups) – see Figures A1 through A4. The results obtain if we change the true value of the parameter. Second, I add another variable into the regression to see if the results hold in multivariate analysis, which they do (Figures A5 and A6).

¹¹ An alternative would be to derive specific predictions from the true model and calculate the bias in each of the predictions. But since we are comparing different modeling techniques, producing baseline probabilities to compare each model requires assumptions that would complicate the estimation and presentation of the results, as well as muddle the intuitiveness of the findings. While the observed probability does not provide an exact measure of bias, it does show that some models are estimated more accurately than others.

Table 1

Datasets used in the simulations. Cells indicate the number of observations per group in each dataset.

Sample Size		1050	2000	2500	3000	3750	4500	5000	7500	100 k
Groups	20		100							
	50			50		75		100		
	75	14								
	100		20		30			50	75	
	150						30			
	1000									100
	4000									25

*Datasets in bold are the focus of the analysis.

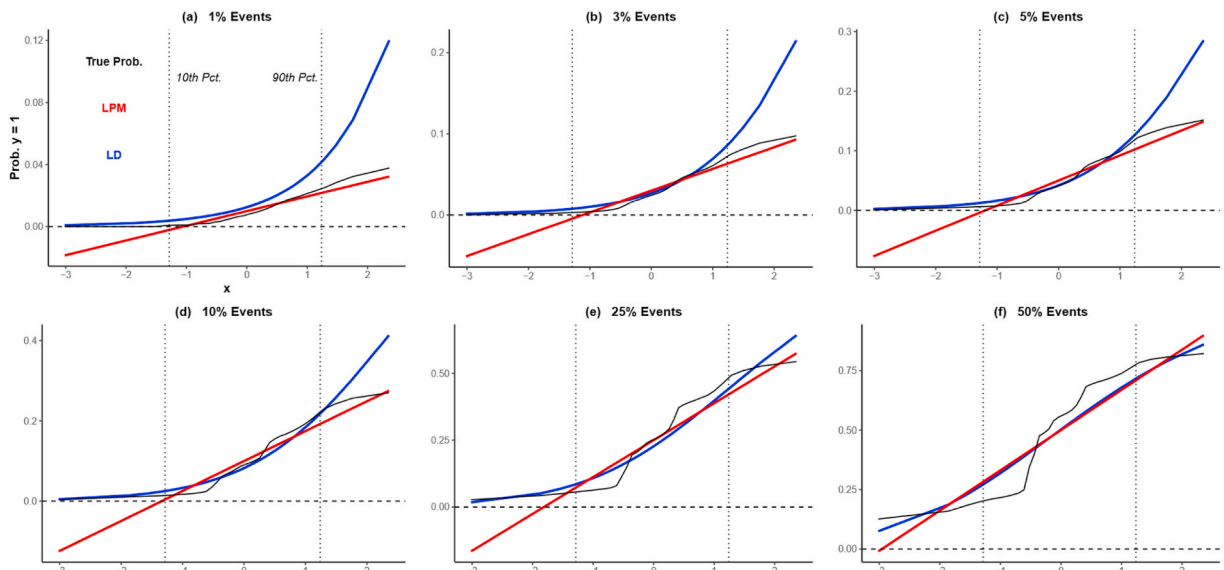


Fig. 2. LPMFE, LD and observed probabilities at different levels of event occurrence in the DV.

probabilities produced by the LPMFE and the LD by the level of occurrence of the event in y , ranging from rare events in (a), where only 1% of the observations are ones, to full variation in plot (f), where there are 50% of ones and 50% of zeros. Presented this way, the results clearly show how the predictions of the LPMFE and the LD evolve as we have more variation in y . The black line –the observed probability– gives us a sense for which model produces more accurate estimates. The line represents the *average* of y in the simulations at different levels of x , or the observed probability in the data. This observed probability gives us an approximation to where the model's prediction should lie –no model ought to produce probabilities that stray far off the actual observed level of probability of y as a function of x .

Three results are particularly important from Fig. 2, and these will apply to Figs. 4 and 5 as well. First, notice that the LPMFE and the ML models produce practically the same predicted probabilities when there is sufficient variation that no groups are dropped in the ML model – see plots (f). The two predictions may differ slightly at the extremes, but they map on to each other perfectly below the 90th percentile and above the 10th.¹² Indeed, this is consistent with what we know about the LPMFE and logit models without fixed effects, that is, that the LPMFE tends to underestimate or overestimate the extremes, but that probabilities in the center of the distribution are similar to those produced by logit models.

Second, between 25 and 75 percent of ones (I only show 25 percent), the ML model performs rather well. The fit of the line is closer to the observed probability, even though the LPMFE model is not far behind. Third, and similarly, what is *much* more striking is the loss of accuracy of the LD predicted probabilities when the number of ones in the data is below 25 percent. Subplots (a) through (d) show that the LPMFE produces probabilities that are much closer the observed probability curve, and virtually all LPMFE lines cross the observed probability curve at the mean. The LD, on the other hand, becomes progressively more inaccurate as events in y become rarer. Thus, these results confirm, on the one hand, that the ML model is highly inaccurate at rare events, which we could reasonably expect

¹² I focus on this range for the interpretation of substantive significance because researchers rarely use min-to-max changes for substantive significance, so the fact that the LPMFE is less accurate at the tails should not be of practical relevance.

given its performance in rare events even without fixed effects. But, on the other hand, they take this common knowledge a step further: inaccuracies in LD predicted probabilities with fixed effects occur even when the data is not rare events per se, but even when y is slightly more common – 3, 5, or 10 percent of events. As plot (d) shows, at 10 percent, even with 100 observations per group, the LD model tends to overestimate the observed probability across the distribution. The LPMFE, on the other hand, maps on nicely to the observed probability when the predictions of the model are over 0. Here, we know that, on average, $y = 1$ is observed around 10 percent on the time in the data. Since the relationship is designed to be positive and statistically significant, we also know that the predictions may be much higher at higher values of x , and be closer to 0 at lower values of x . This is precisely what the observed probability shows and what the LPMFE reflects.

A third important feature of these results is that, at true rare events (that is, around 1 percent of ones or less), the LPM is highly accurate in relation to the observed probability. Intuitively, this makes sense: if we imagine the CDF of the logistic distribution, there are three levels at which we can expect the LPM to practically map onto the ML probabilities *a priori*: at the two extremes and in the middle. This reasoning is illustrated in Fig. 3. The dashed red line represents a linear prediction at rare events, high frequency events, and at the center of the distribution. These lines correspond exactly to their respective parts of the logistic CDF. As the figure illustrates, the linearity assumption imposed by the LPM is less of a concern in rare events. This may appear counterintuitive, as we often consider the CDF of a logistic distribution to concentrate most non-linearities at both ends of the distribution. However, at true rare events, the line is flatter and the model should behave, for most values of x , similarly to the center of the distribution where the probability is closer to fifty percent.¹³

This is, in fact, consistent with the results in Fig. 2. If we look at plots (a) through (f), the black line flattens in comparison to the other plots. The LPMFE models produce probabilities that are much closer to the observed probability of y throughout the range of x . Since the range of predicted probabilities is much smaller in true rare events, say between near-zero probability and 3 or 4 percent, the ML assumption that the effect of x is greater in the middle of the distribution than at the extremes will not produce clear nonlinear patterns, as it may when the range of probabilities is wider. If we could ever fit an LD model and keep all the groups, the probabilities for the LD and the LPMFE would map onto each other the same way they do in plot (m) – at 50 percent of groups with 100 observations per group. Therefore, for fixed effects models at true rare events (below 1 percent) or highly uncommon events (below 5 percent), the LPMFE should be the model of choice. Its positive predicted probabilities will be meaningful and accurate. Lastly, it is important to note that, as expected, the LPMFE produces below-zero predictions in a majority of the plots, but those only occur when both the LD and the observed probabilities are essentially zero.

LPM vs. LD with 30 observations per group (3000 N; 100 G).

Notice also in Figs. 4 and 5 that the LPMFE models produce practically identical probabilities *regardless* of the number of observations or observations per group. Fig. 4 reports the Monte Carlo results using 3000 observations and 100 groups, and Fig. 5 for a dataset with 2000 observations and 100 groups. (The structure of the plots and subplots is the same as in Fig. 2). Some bias may exist in plots between rare events and common events due to the linearity assumption, but the mean prediction produced is usually at the mean of the observed probability of y .

On the other hand, the LD probabilities become increasingly inaccurate as events in y become less common. The robustness of OLS to these problems in model specification are well known, but these results seem to tilt the debate between LPMFE and ML in fixed effects toward the LPMFE. The results are practically identical in both Figures.

LPM vs. LD with 20 observations per group (2000 N; 100 G).

Plots (f) in Figs. 4 and 5 confirm the results from Fig. 2 that, with over 30 observations, both models produce practically the same probabilities if the level of occurrence of y is around 50 percent. Even at 25 observations per group, however, the LD quickly becomes biased, as plots (e) show. At rare events, the LPMFE fits a model that maps on very closely to the observed probability (see plots (a)), again confirming what we saw in Fig. 2. What is important to note is that these datasets are relatively common in comparative politics. It is rather *uncommon* to have much more than 30 observations per group, which is the threshold needed for the LD to produce generally reliable probabilities. In these cases, the ML with fixed effects will be biased even if we manage to obtain data for a large number of countries, say 150. While the n of this dataset would be 3000 or more, the LPMFE should be preferred regardless of y 's composition because the *average* number of observations per group is below 30.

Lastly, Fig. 6 displays the predicted probabilities from both the LPMFE and the LD with big data with rare events – the data in both simulations includes only 1 percent of ones in the dependent variable. The big data results are consistent with the rest of the data structures (Figs. 2, 4 and 5). Here, the most important result is that the LPMFE continues to provide accurate predicted probabilities within the relevant range of the distribution of x . Again, this is expected of the LPMFE: it fits an accurate line through the center of the distribution but fails at the tails.

However, this issue is substantively minor. On the one hand, negative probabilities can be effectively assumed to be zero, and probabilities at high values of x should not generally be used to assess substantively significant effects. The LD, on the other hand, does *better* in big data than in regular comparative dataset structures, in that it produces increasingly positive probabilities that, while still wrong, lie closer to the observed probability than they did in Figs. 2, 4 and 5.

Three conclusions can be derived from this discussion. (1) The LD produces its most accurate predictions between 25 and 75 percent of ones in the dependent variable, and does best when variation in y is highest (50 percent of ones). The LD then becomes

¹³ Note that this is a theoretical argument and that it assumes, for instance, that predictions for the linear probability model to be constrained above zero and below one. While we know that is not the case in practice, the Figure communicates the theoretical intuition that with rare events, the LPM will produce probabilities closer to the logistic regression model than at any other level of frequency in y .

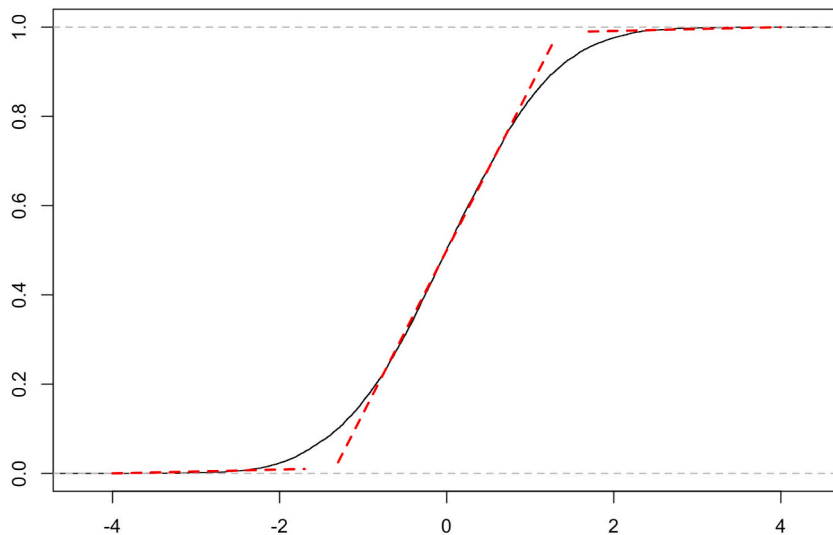


Fig. 3. Logistic CDF with theoretical LPM probabilities overlaid.

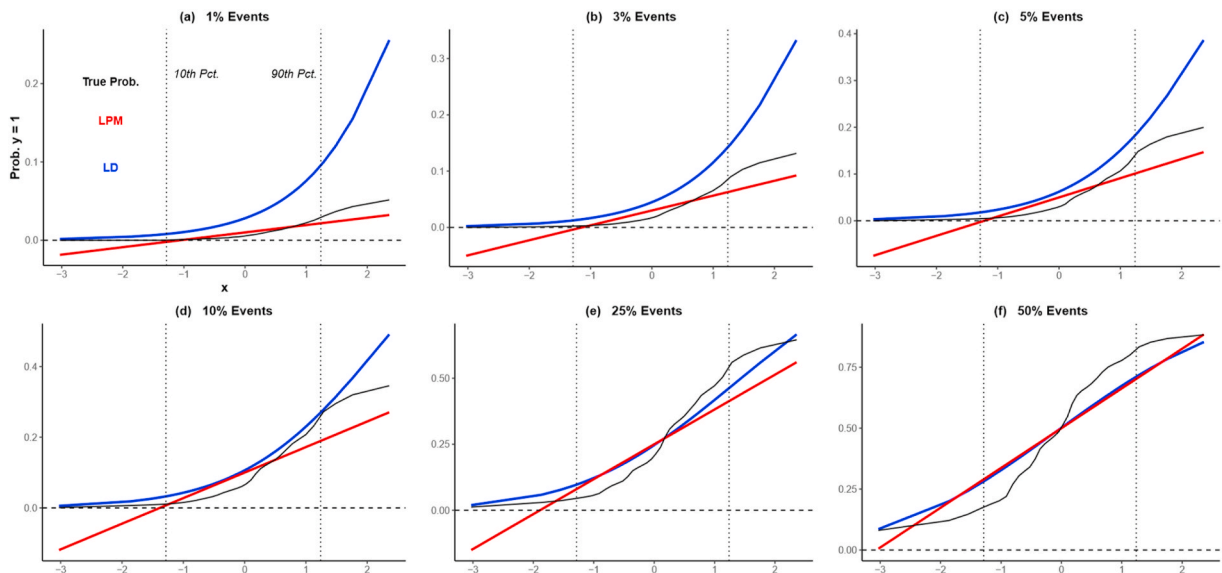


Fig. 4. LPMFE, LD and observed probabilities at different levels of event occurrence in the DV.

increasingly biased as variation in y decreases. (2) The LPMFE fits accurate predictions in true rare events or highly uncommon events data for the part of the distribution that is substantively relevant. (3) Results using big data with rare events remain substantively the same.

One final point to address regarding the simulation results is that the simulation design, as constructed here, assumes that ones are randomly assigned to groups and thus uniformly distributed. This is often not the case in practice, as most datasets have at least some groups with a few ones and others with only one. My set up is certainly the most *favorable* one for logistic specifications, as it leads to the fewest number of observations and groups being removed from the estimation sample. In practice, researchers should first analyze the distribution of ones by group in the data and determine whether my findings are likely to overestimate the performance of the models. We know OLS is relatively robust to a low number of observations and groups, while logistic regression is not. Therefore, if ones are unevenly distributed in the data, logistic specifications should perform much worse than in the simulations, and researchers should take this into account when reading the results and applying them in their own work. Lastly, if the distribution of ones in the data is highly skewed, with most ones concentrated in only a few groups, the researcher should decide whether using fixed effects is appropriate for theoretical reasons, as conclusions for the entire sample would be drawn from the variation of very few groups.

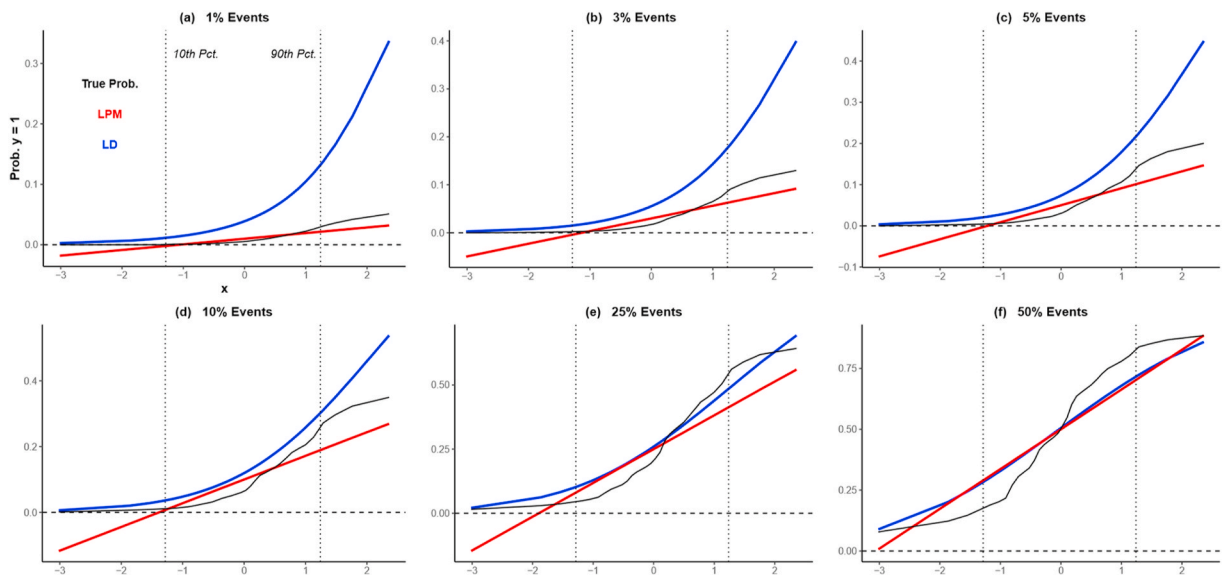


Fig. 5. LPMFE, LD and observed probabilities at different levels of event occurrence in the DV.

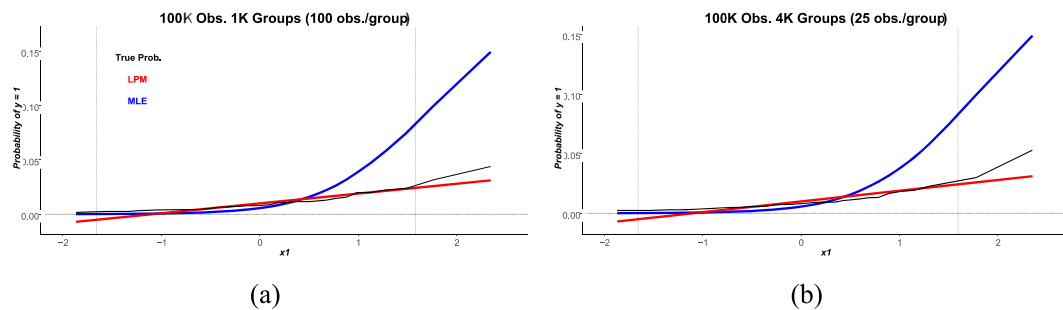


Fig. 6. LPMFE, LD and observed probabilities at different levels of event occurrence in the DV, with big data.

7. Conclusion

TSCS data is a common dataset structure in the social sciences. Repeated observations across time (monthly, quarterly, or yearly data) are pooled for many different groups, such as individuals, districts, states, or countries. Two issues generally emerge with this type of data structure. First, unit-specific effects can lead to biased results if the model does not account for unobservable heterogeneity. Second, theories are often built on domestic processes that should leverage within-group variation, but researchers use between-group variation instead. Fixed effects addresses both of these issues, reducing type-I errors resulting from biased coefficients and using within-group variation in the model. In TSCS data, however, fixed effects estimation can be burdensome when the DV is dichotomous. The dichotomous nature of the DV calls for ML estimation, but a host of issues emerge when using the LD and the CL, as identified in this paper. Interpreting the substantive significance of CL coefficients is often challenging. The LPMFE is the best choice when dealing with rare events or infrequent data among the alternatives analyzed in this article, but the linearity assumption it imposes may not be always justified. Applied researchers are thus placed between a rock and a hard place when trying to use fixed effects models in their research.

In this paper, I have attempted to offer a way out of this dilemma. I have shown that the structure of the dataset is not sufficient to determine which fixed effects model to use. Most analysis of bias in fixed effects used the total number of observations and the number of observations per group to determine the usefulness of ML techniques over the LPMFE. This paper has demonstrated the importance of the *frequency* of events in y to determine model choice. ML increasingly overstates predicted probabilities as events become more rare. The LPMFE, on the other hand, is very accurate at both rare events and highly common events. The same results apply to big data.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ssresearch.2020.102486>.

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